

Alexander Fu

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EDUCATION

University of Waterloo

Honours, Bachelor of Computer Science, Co-op Program

Waterloo, ON

Sept. 2021 – Present

EXPERIENCE

Privacy Engineer Intern

Shopify

Sep. 2025 – Dec. 2025

Toronto, ON

- During a job load anomaly, quickly developed and implemented a **Rails webhook** that automatically locks privacy jobs during critical incidents, projected to save **\$100,000 annually in Google Data Catalog costs**.

Software Engineer Intern

Mesh Connect

May. 2025 – Aug. 2025

San Francisco, CA

- Built and optimized **payment orchestration APIs** enabling global **crypto transactions**, secure **data access**, and **account aggregation** across financial platforms, driving a **30% increase** in successful transfers, a **40% reduction** in cross-platform latency, and processing over **\$120M in annual payment volume**.
- Built a sandbox integration testing environment to simulate payment flows using **Behavior-Driven Development (BDD)** and **Reqnroll**, which cut down manual QA time by **50%** and caught **70+ edge case bugs** prior to production releases.

Software Engineer Intern

Magnet Forensics

Sep. 2024 – Dec. 2024

Waterloo, ON

- Deployed **10+** data pipelines to retrieve suspect cloud data from **WhatsApp** and **Samsung** platforms, in **Golang** and **C#**, containerized and deployed using **Docker**
- Refactored **20+** legacy **E2E test suites** with **Selenium Webdriver**, and **xUnit**, integrating them into our nightly automation testing and coverage analysis using **Jenkins**
- Led development on an **internal artifact generator** to automate the production of containerized artifacts, including documentation and test results with **C#**, **Docker** and **Azure Container Registry (ACR)**

Software Engineer Intern

Miovision

Jan. 2024 - Apr. 2024

Waterloo, ON

- Implemented the **Dynamic A*** algorithm in **Typescript**, enabling adaptive rerouting in case of accidents/traffic
- Developed real-time traffic data ingestion and analytics streams, used in **68** countries and **17,000+** municipalities, leveraging **AWS Kinesis** for high-velocity data streaming and **DynamoDB** for scalable storage
- Engineered **Convolutional Block Attention Modules (CBAMs)** for Miovision's **Convolutional Neural Network (CNN) computer vision model**, improving bound detection success rate by **40%**
- Designed and maintained **15+** scalable, serverless image processing pipelines using **AWS S3** and **Lambda Functions**, to increase traffic data processing speed by **35%**

Software Engineer Intern

BlueStreak Electronics

Apr. 2023 - Aug. 2023

Vaughan, ON

- Built a **Product Catalog System** with **Springboot** and **Angular**, streamlining business logic for autoparts
- Created and maintained **10+ REST APIs** containing auto part data, and refactored existing **APIs** for scalability, reducing load times by up to **10s**

PROJECTS

WingmanAI | AWS, Python, Flask, React, NGINX

<https://github.com/a22fu/WingmanAI>

- Developed a full stack LLM assistant using **LangChain**, **React** and **AWS Bedrock**, deployed on **AWS EC2** with **NGINX** to resolve complex Valorant related queries
- Automated Valorant player data ingestion and transformation using **AWS Glue**, **Athena**, and **S3**, creating a seamless **ETL pipeline** launched on a **Flask API**
- Overcame challenges with model accuracy by integrating a layered approach to data analysis, using **Vector stores** and **Retrieval-Augmented Generation (RAG)** to refine model responses

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, C++, C, Python, Java, HTML/CSS, VB, Go, SQL, R, Kotlin, Ruby on Rails

Tools: AWS Tools, AngularJS, React, Express.js, Docker, Kubernetes, Spring boot, Pandas, Cypress, Nginx, Azure